Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering (An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University) Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu
Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish a learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem solving skills through emerging technologies.

Session 2025-26

Vision: Dream of where you want. **Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and knowby the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for the entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

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Srushti Chaudhari

Name and Signature of Student and Date

(Signature and Date in Handwritten)

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Session	2025-26 (EVEN)	**Course Name**	Artificial Intelligence and Machine Learning
Semester	6	**Course Code**	23CT1606
Roll No	23	**Name of Student**	Srushti Chaudhari

Practical Number	7
Aim	To perform clustering using the K-Means algorithm and analyze the grouping of data points.
Problem Definition	The objective of this practical is to implement the K-Means clustering algorithm to group similar data points into clusters. The algorithm will partition the dataset into a predefined number of clusters (K) based on similarity between data points. The performance of clustering will be analyzed using visualization and cluster grouping.
Theory (100 words)	K-Means Clustering is an unsupervised machine learning algorithm used to group similar data points into clusters. Unlike classification algorithms, K-Means does not use labeled data. The algorithm divides the dataset into K clusters, where each data point belongs to the cluster with the nearest mean (centroid). Key Concepts ● Cluster: A group of similar data points. ● Centroid: The center point of a cluster. ● Distance Measure: Usually Euclidean distance is used to measure

	similarity between points. Working of K-Means 1. Select the number of clusters (K). 2. Initialize K centroids randomly. 3. Assign each data point to the nearest centroid. 4. Recalculate the centroid of each cluster. 5. Repeat steps 3 and 4 until centroids no longer change. Advantages ● Simple and easy to implement ● Works well with large datasets ● Efficient and fast Disadvantages ● Requires selecting the value of K beforehand ● Sensitive to outliers ● Different initial centroids can give different results
Procedure and Execution (100 Words)	**Code**: # Import libraries import pandas as pd import matplotlib.pyplot as plt from sklearn.cluster import KMeans # Load dataset

```
# Assuming 'iris.csv' is in your directory
data = pd.read_csv("iris.csv")

# Select features (e.g., Sepal Length and Sepal Width)
# .values converts the dataframe slice to a NumPy array for
easier plotting
X = data.iloc[:, [0, 1]].values

# Apply K-Means
# n_init='auto' is recommended for compatibility with
scikit-learn 1.2+
kmeans = KMeans(n_clusters=3, n_init='auto',
random_state=42)

# Fit and Predict in one step
y_kmeans = kmeans.fit_predict(X)

# Plot clusters
plt.figure(figsize=(8, 6))

# Plotting the data points, colored by their assigned cluster
plt.scatter(X[:, 0], X[:, 1], c=y_kmeans, cmap='viridis',
label='Data Points')

# Plotting the Centroids (the 'X' marks)
plt.scatter(kmeans.cluster_centers_[:, 0],
kmeans.cluster_centers_[:, 1],
            s=200, c='red', marker='X', label='Centroids')

plt.title("K-Means Clustering on Iris Dataset")
plt.xlabel("Sepal Length (cm)")
plt.ylabel("Sepal Width (cm)")
plt.legend()
plt.grid(True, linestyle='--', alpha=0.6)
plt.show()
```

Github link -	Output: Cluster Labels (first 10 data points): [1 1 1 1 1 1 1 1 1 1] Cluster Centers: [[5.80 2.70] [5.00 3.40] [6.90 3.10]] https://github.com/SRchaudhari15/AIML-Practicals.git
Output Analysis	● The dataset is divided into 3 clusters. ● Each data point is assigned a cluster label (0,1,2). ● The cluster centers represent the mean position of points in each cluster. ● Data points belonging to the same cluster have similar feature values.
Conclusion	In this practical, the K-Means clustering algorithm was implemented successfully to group data points into clusters. The algorithm identified patterns in the dataset without using labeled data and grouped similar observations together.
Date	.2/3/2026